DE–6201 **11**

DISTANCE EDUCATION

M.Sc.(Maths) DEGREE EXAMINATION, MAY 2013.

ALGEBRA

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) If G is a finite group and $a \in G$ then prove that O(a)/ O(G). (10)

   (b) Prove that the sub group N of G is a normal sub group of G if and only if every left coset of N in G is right coset of N in G. (10)

2. (a) Prove that if G is a group, then A(G), the set of Automorphisms of G is also a group. (10)

   (b) Prove that the number of p – sylow subgroups in G, for a given prime is of the form 1 + Kp. (10)

3. (a) Let G be a group and suppose that G is the internal direct product of N_1, N_2....., N_n Let T = $N_1 \times N_2 \times \ldots\ldots \times N_n$. Then prove that G and T are isomorphic. (10)

   (b) Prove that two abelian groups of order p^n are isomorphic if and only if they have the same invariants. (10)

4. (a) Let R be a commutative ring with unit element whose only ideals are (0) and R itself. Then prove that R is a field. (10)

   (b) If R is a unique factorization domain then prove that So is R[x]. (10)

5. (a) If V is the internal direct sum of V_1, V_2.....V_n, then prove that V is isomorphic to the external direct sum of V_1, V_2.....V_n. (10)

(b) If V_1, V_2.....V_n is a basic of V over F and if w_1, w_2,.....w_m in V are linearly independent over F, then prove that $m \leq n$. (10)

6. (a) If $T \in A$ (V) has all its characteristics roots in F, then prove that there is a basis of V in which the matrix of T is triangular. (10)

(b) For each $i = 1, 2, ...K, V_1 \neq (0)$ and $V = V_1 \oplus V_2 \oplus ..... \oplus V_K$. Then prove that the minimal polynomial of T_i is $q_i(x)^{li}$. (10)

7. (a) Prove that the element $a \in K$ is algebraic over F if and only if F(a) is a finite extension of F. (10)

(b) If P(x) is a polynomial in F[x] of degree $n \geq 1$ and is irreducible over F, then prove that there is an extension E of F, such that [E : F] = n, in which p(x) has a root. ss (10)

8. (a) If F is a field of characteristics $p \neq 0$, then prove that the polynomial $x^{px} - x \in F[x]$, for $n \geq 1$, has distinct roots. (10)

(b) Let K be a normal extension of F and let H be a subgroup of G(K,F) let

$K_H = \{x \in K / \sigma(x) = x \text{ for all } \sigma \in H\}$ be the fixed field of H. Then prove that

(i) $[K : K_H] = O(H)$

(ii) $H = G(K, K_H)$. (10)

DE–6202 **12**

DISTANCE EDUCATION

M.Sc.(Maths) DEGREE EXAMINATION, MAY 2013.

REAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Prove that every K-cell is compact. (10)

   (b) If X is a metric space and $E \subset X$ then prove that

   (i) $\overline{E}$ is closed

   (ii) $E = \overline{E}$ if and only if E is closed. (10)

2. (a) Suppose f is continuous real function on a compact metric space X and

   $M = \sup_{p \in X} f(p)$ $\quad$ $m = \inf_{p \in X} f(p)$

   Then prove that there exist $p, q \in X$ such that $f(p) = M$ and $f(q) = m$. (10)

   (b) Suppose f is a continuous 1-1 mapping of a compact metric space X onto a metric space Y. Then prove that the inverse mapping f^{-1} defined on Y by $f^{-1}(f(x) = x\,(x \in X)$ is a continuous mapping of Y onto X. (10)

3. (a) Suppose f is continuous on $[a,b], f'(x)$ exists at some point $x \in [a,b], g$ is defined on an interval I which contains the range of f and g is differentiable at the point $f(x)$. If $h(t) = g(f(t))$ $(a \le t \le b)$ then prove that h is differentiable at x and $h'(x) = g'(f(x))(f'(x))$ (10)

(b) Let f be defined on [a,b]; if f f has a local maximum at a point $x \in (a,b)$ and if $f'(x)$ exists then prove that $f'(x) = 0$. (10)

4. (a) Suppose f is a continuous mapping of $[a,b]$ into R^k and $\bar{f}$ is differentiable in (a,b) Then prove that there exists $x \in (a,b)$ such that $\left|\bar{f}(b) - \bar{f}(a)\right| \le (b-a)\left|f'(x)\right|$. (10)

(b) Suppose f is a l mapping of an open set $E \subset R^n \, into \, R^n, f'(a)$ is a invertible for some $a \in E$ and $b = f(a)$ Then prove that there exist open sets U and V in R^n such that $a \in U, b \in V, f$ is one to one on U and $f(U) = V$. (10)

5. (a) If P* is a refinement of P then prove that $L(P,f,\alpha) \le L(P^y,f,\alpha)$ and $U(P,f,\alpha) \le U(P,f,\alpha)$. (10)

(b) State and prove the fundamental theorem of calculus. (10)

6. (a) If $f \in R(\alpha)$ and $g \in R(\alpha)$ on $[a,b]$ then prove that

(i) $f\,g \in R(\alpha)$

(ii) $|f| \in R(\alpha)$ and $\left|\int_a^b f d\alpha\right| \le \int_a^b |f|\, d\alpha$ (10)

(b) Suppose ϕ is a strictly increasing continuous function that maps an $[A,B]$ *onto* $[a,b]$ suppose α is monotonically increasing on [a,b] and $f \in R(\alpha)$ on [a,b]. Define β and g on [A,B] by $\beta(y) = \alpha(\phi(y), g(y) = f(\phi(y)$. Then prove that $g \in R(\beta)$ and $\int_A^B g d\beta = \int_a^b f d\alpha$. (10)

7. (a) Let $< E_n >$ be an infinite decreasing sequence of measurable sets, that is a sequence with $E_{n+1} \subset E_n$ for each n. Let mE_1 be finite Then prove that $m\left(\bigcap_{i=1}^{\infty} Ei\right) = \lim_{n \to \infty} mE_n$. (10)

(b) State and prove $Egoroff's$ Theorem. (10)

8. (a) State and prove fatou's lemma. (10)

(b) State and prove Lebesgue Convergence Theorem. (10)

DE–6203 **13**

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2013.

DIFFERENTIAL EQUATIONS AND NUMERICAL METHODS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) If $y_1(x)$ and $y_2(x)$ are two solutions of equation $y'' + p(x)y' + \varphi(x)y = R(x)$ on $[a,b]$, then they are linearly dependent on this interval iff their Wronskian $W(y_1, y_2) = y_1 y_2' - y_2 y_1'$ is identically zero.

   (b) Show that $y = c_1 \sin x + c_2 \cos x$ is the general solution of $y'' + y = 0$ on any interval, and find the particular solution for which $y(0) = 2$ and $y'(0) = 3$.

2. (a) Find the general solution of $y'' + 3y' - 10y = 6e^{4x}$.

   (b) Express $\sin^{-1} x$ in the form of a power series $\sum anx^n$ by solving $y' = \left(1 - x^2\right)^{-\frac{1}{2}}$ in two ways (Hint : remember the binomial series). Use this result to obtain the formula $\frac{\pi}{6} = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{3.2^3} + \frac{1.3}{2.4} \cdot \frac{1}{5.2^5} + \frac{1.3.5}{2.4.6} \cdot \frac{1}{7.2^7} + \ldots$

3. Solve the equation $\frac{\partial^2 z}{\partial x^2} - 2\frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} = 0$, by the method of separation of variables.

4. The ends of a uniform string of length $2l$ are fixed. The initial displacement is $y(x,0) = kx(2l - x)$, $0 < x < 2l$, while the initial velocity is zero. Find the displacement at any distance x form the end $x = 0$ at any time t.

5. Find the real root lying between 1 and 2 of the equation $x^3 - 3x + 1 = 0$ upto 3 places of decimals by using Regula–falsi method and bisection method.

6. Solve the least–squares approximation problem of $f_n \approx f(x_n)$ with $x_n = 10 + \frac{n-1}{5}$, $n = 1, 2, \ldots\ldots, 6$. Using orthogonal polynomials.

7. Evaluate $\int_0^1 \frac{dx}{1+x}$ using

   (a) Trapezoidal rule

   (b) Simpson's one third rule

   (c) Simpson's three eight rule.

8. Find the solution of the differential equation $y' = 1 - y, y(0) = 0$ at $x = 0.1$ and $x = 0.2$ correct to four decimal places.

DE–6204 **14**

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2013.

OPERATIONS RESEARCH

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Solve the following LPP by graphically. (10)

Maximum $z = 5x_1 + 7x_2$

Subject to $x_1 + x_2 \le 4$

$3x_1 + 8x_2 \le 24$

$10x_1 + 7x_2 \le 35$ and

$x_1, x_2 \ge 0$.

(b) Solve the following LPs by the revised simplex method. (10)

Maximize $Z = 2x_1 + x_2 + 2x_3$

Subject to $4x_1 + 3x_2 + 8x_3 \le 12$

$4x_1 + x_2 + 12x_3 \le 8$

$4x_1 - x_2 + 3x_3 \le 8$

$x_1, x_2, x_3 \ge 0$.

2. (a) Explain sensitivity analysis. (10)

(b) Find the integer solution to the LPP (10)

Maximize $Z = 2x_1 + 2x_2$

Subject to the constraints

$5x_1 + 3x_2 \le 8$,

$x_1 + 2x_2 \ge 4$ and $x_1, x_2 \ge 0$ and are integers.

3. (a) Let a_{ij} be the (i, j)th element of a payoff matrix with m strategies for player A and n strategies for player B. The payoff is for player A. Prove that (10)

$$\max_i \min_j a_{ij} \le \min_j \max_i a_{ij}$$

(b) Using dynamic programming approach, solve the reliability problem with the following data : (10)

K_j	$j=1$		$j=2$		$j=3$	
	R_1	C_1	R_2	C_2	R_3	C_3
1	0.6	2	0.8	3	0.7	1
2	0.7	4	0.8	5	0.8	2
3	0.9	5	0.9	6	0.9	3

The total capital available is 10(in units of thousand rupees)

4. (a) Specify the range for the value of the game in each of the following cases assuming that the payoff is for player A: (10)

	B_1	B_2	B_3	B_4
A_1	1	9	6	0
A_2	2	3	8	4
A_3	–5	–2	10	–3
A_4	7	4	–2	–5

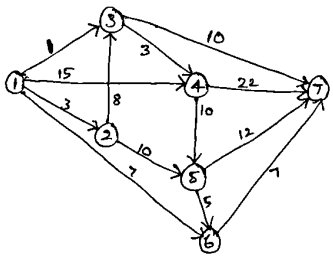
(b) Solve the following game graphically. The payoff is for player A. (10)

	B_1	B_2	B_3
A_1	1	–3	7
A_2	2	4	–6

5. (a) (i) Define network

(ii) How many rules are available for constructing the network? and explain. (10)

(b) Determine the critical path for the following project network (10)



6. (a) The probability distribution of the demand for a certain items is as follows : (10)

Monthly sales :	0	1	2	3	4	5	6
Probability :	0.01	0.06	0.25	0.35	0.20	0.03	0.10

The cost of carrying inventory is Rs. 30/unit per month and the cost of unit short is Rs. 70/month. Find the optimum stock level which will maximize the total expected cost.

(b) Construct the project network comprised of activities A to P that satisfies the following precedence relationships. (10)

(i) A, B and C, the first activities of the project, can be executed concurrently.

(ii) D, E and F follow A

(iii) I and G follow both B and D

(iv) H follows both C and G

(v) K and L follow I

(vi) J succeeds both E and H

(vii) M and N succeed F, but cannot start until both E and H are completed

(viii) O succeeds M and I

(ix) P succeeds J,L and O

(x) K, N and P are the terminal activities of the project.

7. (a) The time between arrivals at the game room in the student union is exponential with mean 10 minutes. (10)

(i) What is the arrival rate per hour?

(ii) What is the probability that no students will arrive at the game room during the next 15 minutes?

(iii) What is the probability that atleast one student will visit the game room during the next 20 minutes?

(b) Explain single server models. (10)

8. (a) For the $(M/M/I):\ (GD/\infty/\infty)$ show that (10)

(i) The expected number in the queue given that the queue is not empty = $\left(\frac{1}{(1-\rho)}\right)$.

(ii) The expected waiting time in the queue for those who must wait $= \frac{1}{(\mu-\lambda)}$.

(b) For the $(M/M/C):\ (GD/\infty/\infty)$ model, show that (10)

(i) The probability that a customer is waiting is $\left\{\frac{\rho}{(c-\rho)}\right\}P_c$.

(ii) The average number in the queue given that it is not empty is $\frac{c}{(c-\rho)}$.

(iii) The expected waiting time in the queue for customers who must wait is $\frac{1}{\mu(c-\rho)}$.

DE–6205 **15**

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2013.

MATHEMATICAL STATISTICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks. $(5 \times 20 = 100)$

1. (a) Given : $f(x,y)=e^{-(x+y)} I_{(0,\infty)}(x).I_{(0,\infty)}(y)$. Are X and Y independent? Find

   (i) $P(X>1)$,

   (ii) $P(X<Y/X<2Y)$

   (iii) $P(1<X+Y<2)$. (10)

   (b) Two ideal dice are thrown. Let X, be the score on the first die and X_2 be the score on the second die. Let Y denote the maximum of X_1 and X_2, (ie) $\gamma=\max(X_1,X_2)$.

   (i) Write down the joint distribution of Y and X_1 and

   (ii) Find the mean and variance of Y and covariance (Y, X_1). (10)

2. (a) Two tetrahedral with sides numbered 1 to 4 are tossed. Let X denote the number on the downturned face of the first tetrahedron and Y denote the larger of the downturned numbers. Investigate the following:

   (i) Joint density function of X,Y and marginals f_x and f_y.

   (ii) $P\{X\leq 2, Y\leq 3\}$,

(iii) $Cov(X,Y)$

(iv) $E(Y/X=2)$

(v) Construct joint density different from that in part (a) but processing same marginals f_X and f_Y. (10)

(b) If X is a Gamma variate with parameter λ, obtain its m.g.f. Hence deduce that the m.g.f of standard gamma variate tends to exp ($\frac{1}{2}t^2$) as $\lambda \to \infty$. Also interpret the result. (10)

3. (a) Let $X_1, X_2, \ldots X_n$ be jointly normal with $E(X_i)=0$ and $E(X_i^2)=1$ for all i and $\text{cov}(X_i, X_j)=e$ if $|j-i|=1$ and = 0, otherwise. Examine if WLLN holds for the sequence $\{X_n\}$. (10)

(b) If the random variables X_1 and X_2 are independent and follow chi-square distribution with n d.f., show that $\sqrt{n}\,(X_1-X_2)/2\sqrt{X_1X_2}$ is distributed as student's t with n d.f., independently of X_1+X_2. (10)

4. (a) State and prove weak law of large numbers. (10)

(b) A random sample of 10 boys had the following I.Q.'s : 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100? Find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie. (10)

5. (a) A machine puts out 16 imperfect articles in a sample of 500 articles. After the machine is overhauledit puts out 3 defective articles in a sample of 100. Has the machine improved? (10)

(b) Explain the concepts of Type I and Type II errors, with examples. (10)

6. State and prove Neyman-Pearson lemma. (20)

7. (a) Write down the applications of C-chart. (10)

(b) From a lot consisting of 2,200 items, a sample of size 225 is taken. If it contains 14 or less defectives, the lot is accepted otherwise rejected. Plot OC, ATI and AOQ curves. Also obtain the value of A.O.Q.L. (10)

8. (a) State the mathematical model used in analysis of variance in a two-way classification. Explain the hypothesis to be used. Discuss the advantages of this method. (10)

(b) The following table gives quality rating of service stations by five professional rates:

Rater	1	2	3	4	5	6	7	8	9	10
A	99	70	90	99	65	85	75	70	85	92
B	96	65	80	95	70	88	70	51	84	91
C	95	60	48	87	48	75	71	93	80	93
D	98	65	70	95	67	82	73	94	86	80
E	97	65	62	99	60	80	76	92	90	89

Analyse the data and discuss whether there is any significance difference between ratings or between service stations. (10)

DE–6206 **21**

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2013.

COMPLEX ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$ marks)

1. (a) Let $z = r(\cos\theta + i\sin\theta)$ be any non-zero complex number and $"n"$ by any integer. Then prove that $z^n = r^n(\cos n\theta + i\sin n\theta)$. (10)

   (b) If $u + v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$. Find the analytic function $f(z)$ in terms of z. (10)

2. (a) Let f be an analytic function defined in a region D. Ltd $Z_0 \in D$. If $f'(z_o) \neq 0$ then prove that f is conformal at z_0. (10)

   (b) Find the image of the circle $|z - 3i| = 3$ under the map $w = \frac{1}{z}$. (10)

3. (a) State and prove cauchy's theorem for multiply connected regions. (10)

(b) Show that

(i) $\frac{1}{z^2} = 1 + \sum_{n=1}^{\infty} (n+1)(z+1)^n$ when $|z+1| < 1$. (10)

(ii) $\frac{1}{z^2} = \frac{1}{4} + \frac{1}{4}\sum_{n=1}^{\infty} (-1)^n (n+1)\left(\frac{z-2}{2}\right)^n$ when $|z-2| < 2$.

4. (a) State and prove morera's theorem. (10)

(b) Suppose the $\phi(\varepsilon)$ is continuous on the arc γ. Prove that the function $F_n(z) = \int_{\gamma} \frac{\phi(\varepsilon)d\varepsilon}{(\varepsilon - z)^n}$ is analytic in each of the regions determined by γ and its derivative is given by $F_n^{'}(z) = n\, F_{n+1}(z)$. (10)

5. (a) Let $f(z)$ be a function having "a" as an isolated singular point. Then prove that the following are equivalent. (10)

(i) "a" is a pole of order "r" for *f(z)*.

(ii) *f(z)* can be written in the form $f(z) = \frac{1}{(z-a)^r}\theta(3)$ where $\theta(z)$ has a removable singularity at $z = a$ and $\lim_{z \to a} \theta(z) \neq 0$.

(iii) "a" is a zero of order r for $\frac{1}{f(z)}$.

(b) Prove that $\int_{O}^{\infty} \frac{x^4 dx}{x^6 - 1} = \frac{\pi\sqrt{3}}{6}$. (10)

6. (a) State and prove Jensen's formula. (10)

(b) State and prove legendre's duplication formula. (10)

7. (a) Prove that a discrete module consists either of zero alone of the integral multiples nw of a single complex number $w \neq 0$, or of all linear combinations $n_1 w_1 + n_2 w_2$ with integral coefficients of two numbers w_1, w_2 with nonreal ratio w_2 / w_1. (10)

(b) Derive the first order differential equation for $w = p(3)$. (10)

8. (a) Prove that the zeros $a_1, a_2, \ldots a_n$ and poles $b_1, b_2, \ldots b_n$ of an elliptic function satisfy $a_1 + a_2 + \ldots + a_n \equiv b_1 + b_2 \ldots + b_n$ (mod M). (10)

(b) Prove that the sum of the residues of an elliptic function is zero. (10)

DE–6207 **22**

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2013.

TOPOLOGY AND FUNCTIONAL ANALYSIS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

Each question carries 20 marks.

($5 \times 20 = 100$ marks)

1. (a) Prove that any continuous image of a compact space is compact. (10)

   (b) Prove that every separable metric space is second countable. (10)

2. (a) State and prove Tychonoff's theorem. (10)

   (b) State and prove Lebesgue's covering Lemma. (10)

3. (a) State and prove Urysohn's imbedding theorem. (10)

   (b) State and prove Tietze extension theorem. (10)

4. (a) Prove that continuous image of a connected space is connected. (10)

   (b) Prove that the product of any non-empty class of Hasudorff spaces is Hausdorff. (10)

5. (a) State and prove the Hahn-Banach theorem. (10)

   (b) State and prove the open-mapping theorem. (10)

6. (a) State and prove uniform-boundedness theorem. (10)

   (b) State and prove the closed graph theorem. (10)

7. (a) If T is an operator on H, <Tx, x> = 0, for all x, then prove that T = 0. (10)

(b) If N_1 and N_2 are normal operators on H. Such that either commutes with the adjoint of the other than prove that $N_1 + N_2$ and $N_1 N_2$ are normal. (10)

8. (a) Derive Bessel's inequality. (10)

(b) If A_1 and A_2 are self-adjoint operators on H_1 then prove that $A_1 A_2$ is self adjoint if $A_1 A_2 = A_2 A_1$. (10)

DE–6208 **23**

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2013.

GRAPH THEORY

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Prove that $\delta \leq \frac{2q}{p} \leq \Delta$. (5)

(b) Let G be a (p, q) graph, prove that the following are equivalent.

(i) G is a tree

(ii) Every two points of G are joined by a unique path.

(iii) G is connected and $p = q + 1$

(iv) G is acyclic and $p = q + 1$. (10)

(c) Prove that every connected graph has a spanning tree. (5)

2. (a) Let v be a point of a connected graph G. Prove that the following are equivalent.

(i) v is a cut-point of G.

(ii) There exist a partition of $V - \{v\}$ into subsets U and W such that for each $u \in U$ and $w \in W$, the point v is on every $u - w$ path.

(iii) There exist two points u and w distinct from v such that v is on every $u - w$ path. (10)

(b) Explain Kruskal's algorithm. (10)

3. (a) Prove that a graph G is connected if and only if for any partition of V into subsets V_1 and V_2 there is G line of G joining a point of V_1 to a point of V_2. (10)

(b) Prove that for any graph G, $K \le \lambda \le \delta$. (10)

4. (a) Let G be a graph with p points and let u and v be nonadjacent points in G such that $d(u)+d(v) \ge p$. Prove that G is hamiltonian if and only if $G+uv$ is hamiltonian. (10)

(b) State and prove Chavatal's theorem. (10)

5. (a) Prove that $r(m, n) = r(n, m)$. (5)

(b) Prove that $\alpha + \beta = p$. (5)

(c) For any graph G, prove that $X(G) \le 1 + \max \delta(G')$ where the maximum is taken over all induced subgraphs G' of G. (10)

6. (a) State and prove Brook's theorem. (10)

(b) Prove that the coefficients of $f(G, \lambda)$ alternate in sign. (10)

7. (a) If G is a connected plane graph having V, E and F as the set of vertices, edges and faces respectively, prove that $|V| - |E| + |F| = 2$. (10)

(b) Prove that every planar graph is 5-colourable. (10)

8. (a) If two diagraphs are isomorphic, prove that the corresponding points have same degree pair. (10)

(b) For any flow f and any cut $(s, \bar{s})$ in N prove that $\text{val } f = f^+(s) - f^-(s)$. (10)

DE–6209 **24**

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2013.

PROGRAMMING IN C/C++

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain the characteristics of a object oriented programming. (10)

   (b) Discuss the different types of constants in C programming with examples. (10)

2. (a) Discuss the structure of a C program with example. (10)

   (b) Explain comments in C/C++ and variable declaration with examples. (10)

3. (a) Explain the following functions with examples

   (i) Gets ( )

   (ii) Put char ( )

   (iii) Get char ( ) (10)

   (b) Write a function to find the factorial of a given number. Use it in main ( ) function. (10)

4. (a) Discuss the Arithmetic operators in C programming with examples. (10)

   (b) Explain local and global variables with examples. (10)

5. (a) Discuss the relational and logical operators with examples. (10)

   (b) Explain Nested if statements with an example. (10)

6. (a) Discuss the following looping with examples

   (i) Do…while

   (ii) While

   (iii) For (10)

   (b) Explain the following string functions with examples.

   (i) Strcat

   (ii) Strcmp

   (iii) Strcpy

   (iv) Strlen (10)

7. (a) Discuss the structure declaration with examples.(10)

   (b) Explain structures and functions with an example. (10)

8. (a) Discuss the following with examples (10)

   (i) Opening a file

   (ii) Closing a file

   (iii) Declaring a file

   (b) Write a program to copy the contents of one file into another. (10)

DE–6210 **25**

DISTANCE EDUCATION

M.Sc. (Maths) DEGREE EXAMINATION, MAY 2013.

DISCRETE AND COMBINATORIAL MATHEMATICS

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Let a_n denote the number of n-digit sequences in which each digit is 0 or 1, no two consecutive 0s being allowed. Show that $a_1 = 2, a_2 = 3$ and that $a_n = a_{n-1} + a_{n-2}$ $(n \geq 3)$. Hence find a_n. (10)

(b) Solve the recurrence relation

$a_{n+2} - 6a_{n+1} + 9a_n = 3(2^n) + 7(3^n), n \geq 0$ given that $a_0 = 1$ and $a_1 = 4$. (10)

2. (a) Prove that $\sum_{\phi \in R^D} W(\phi) = \left(\sum_r W(r)\right)^{|D|}$. (10)

(b) If 6 people A, B, C, D, E, F, are selected about a sound table, how many different circular arrangements are possible, if arrangements are considered the same when one can be obtained from the other by rotation? (10)

3. (a) State and prove generalised inclusion and exclusion principle. (10)

(b) Show that the number of dearrangements of a set of n element is given by

$$D_n = n!\left[1 - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + ... + (-1)^n \frac{1}{n!}\right].$$ (10)

4. (a) Show that the number of partitions of n in which no integer occurs more than $(r-1)$ times as a part is equal to the number of partitions of n into parts not divisible by r. (10)

(b) Find the number of integers between 1 and 250 both inclusive that are not divisible by any of the integers 2, 3, 5 and 7. Use principle of inclusion – exclusion. (10)

5. (a) If C_n is the cyclic group of permutations on n symbols then prove that $Z(c_n; s_1, s_2....s_n) = \frac{1}{n}\sum_{d/n} \phi\left(\frac{n}{d}\right) s_{n/d}^d$, where ϕ is the euler function. (10)

(b) State and prove Polya's generalization theorem. (10)

6. (a) If in Polya's theorem (PET 3) the group G is S_n, show that the number of configurations which do not contain a figure more than once is given by polya–substitution of $((x) \text{ in } Z(A_n - S_n))$. (10)

(b) Let ω be any function defined on S satisfying $\omega(gs) = \omega(s)$ for all $g \in G$. Let s be a set of representatives of the G-equivalence classes of S defined by G. Then, for each $i = 1, 2, ...N$, prove that

$$\sum_{s \in 0} w(s)\chi(G_{(s)} \sim G_i) = \sum_{j=1}^{N} C_{ij} \sum_{s \in S} w(s)\chi(G_j < G_{(s)}).$$

(10)

7. (a) Let L be any non-empty set with two binary operations $\vee$ and $\wedge$ defined on it and satisfying $L_1, L_2, L_3, L_4, L_1', L_2', L_3'$ and L_4'. Then prove that L is a lattice relative to a suitable definition of $\leq$ and $\vee$ and $\wedge$ are the l.u.b and g.l.b in this lattice. (10)

(b) Using Karnaugh map, simplify the function

$$f(a, b, c, d) = \sum (0, 2, 6\ 7, 8, 9, 13, 15).$$ (10)

8. (a) Show that in any distributive lattice

$$(a \vee b) \wedge (b \vee c) \wedge (c \vee a) = (a \wedge b) \vee (b \wedge c) \vee (c \wedge a).$$ (10)

(b) Prove that the lattice of normal subgroups of any group is a modular lattice. (10)